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Trading Games: Beating Passive Strategies in the Bullish Crypto Market

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ABSTRACT

This study examines the effectiveness of cointegrated pairs trading in cryptocurrency markets, introducing systematic parameter optimization within the trading framework. The analysis is conducted using a dataset comprising ten major cryptocurrencies, selected based on market capitalization and consensus mechanism, spanning the period from January 2019 to May 2024. The methodology incorporates dynamic risk management through adaptive trailing stop-loss and volatility filtering mechanisms. Empirical results demonstrate that the pairs trading strategy consistently outperforms conventional pairs trading and passive approaches, generating significant risk-adjusted excess returns, while maintaining low market exposure. These findings contribute to the literature on empirical finance and provide valuable insights into alternative investment strategies for emerging digital asset markets.

JEL Classification: C22, C58, G11, G14

1 | Introduction

The search to leverage market inefficiencies to achieve enhanced returns has long occupied a central focus in financial research, with ongoing efforts aimed at refining strategies to maximize portfolio performance. Two prominent approaches that have emerged from this literature are momentum and pairs trading strategies. The former—as elucidated by Jegadeesh and Titman (1993)—demonstrates that a portfolio strategy of buying past winners and selling past losers can yield significant abnormal returns. This suggests that 'momentum' can be attributed to price movements deviating from their long-term values, temporarily causing prices to overreact. The persistence of momentum effects indicates a potential inefficiency in the speed at which markets incorporate information. Chan et al. (1996) further investigate this aspect, examining the market's reaction to information and its relationship to the predictability of future returns. Their findings highlight how the gradual diffusion of information can lead to exploitable market inefficiencies, underpinning the theoretical basis for momentum strategies.

By contrast, pairs trading strategy is a market-neutral approach that involves selecting two cointegrated assets and exploiting the spread between them by selling the overperforming asset (the "winner") and buying the underperforming one (the "loser"). This approach aligns with the concept of statistical arbitrage as outlined by Gatev et al. (2006), who demonstrated its profitability in equity markets. The theoretical foundation of pairs trading relies on the expectation that cointegrated assets will revert to their equilibrium relationship over time, creating arbitrage opportunities. Furthermore, Huck and Afawubo (2015) show that selecting pairs based on cointegration leads to higher and more stable returns compared to alternative selection methods.

However, research on pairs trading strategies remains limited, particularly in the cryptocurrency market. Jacobs and Weber (2015) highlight that although academic literature largely overlooks pairs trading, hedge funds and investment banks widely use it due to its profitability. The scarce literature highlights the opportunities within pairs trading strategies. For example, Gatev et al. (2006) underline that investors with relatively low transaction costs can more effectively capitalize on mispricing, generating profits from market inefficiencies. (Chen et al. 2019) demonstrate that short-term reversals partially explain the returns from pairs trading strategies, while Do and Faff (2010) argue that such strategies exhibit superior performance during prolonged market turbulence and in illiquid bear markets.

In this study, I propose a simple yet novel enhancement to the traditional pairs trading strategy through an optimization procedure for the lookback period using grid-search. In the model, a broad search over a range of possible lookback periods is performed, and as the sample size increases, the grid-search converges to an optimal lookback period that maximizes the in-sample Sharpe ratio. By optimizing the lookback period during the in-sample phase, the model achieves more precise parameter estimation and adapts better to changing market conditions. The process for pair selection is twofold: first, cointegrated pairs are identified; second, among these, the pair with the highest Sharpe ratio is selected and compared against a traditional pairs trading apparoach and a passive buy-and-hold benchmark for its constituent cryptocurrencies. The proposed framework incorporates a dynamic calibration mechanism through a rolling window framework. This approach calculates the time-varying mean and standard deviation of the spread using the optimized lookback period, ensuring continuous parameter recalibration as new market data becomes available. This dynamic adjustment enables the model to incorporate current market conditions rather than relying on static historical parameters. Additionally, I identify and assess strategy performance across different market regimes by applying the Zegadlo (2022) algorithm for detecting bull and bear markets. This algorithm identifies turning points in price trends, enabling performance analysis under varying market conditions.

Since cryptoassets are known for their high volatility, I have implemented a volatility-filtering mechanism that suppresses trading signals when current volatility exceeds 1.5 times the average volatility. This approach helps manage exposure during periods of heightened market turbulence. In addition, the model accounts for practical trading constraints—including transaction costs at both entry and exit—as well as a dynamic trailing stop-loss mechanism.

Shleifer and Vishny (1997) argue that asset managers should avoid extreme volatility because investors might pressure them to liquidate positions that incur severe losses, potentially damaging their reputation. One way to mitigate this risk is by implementing a stop-loss mechanism. Kaminski and Lo (2014) highlight how behavioral biases—such as "selling at the low" and "buying at the high," particularly during market turmoil—can lead to suboptimal decision-making. Their research shows that appropriately designed stop-loss policies can not only generate positive stopping premia but also help reduce volatility

The stop-loss mechanism is designed as an automated tool that limits losses by exiting positions (through either selling or buying) when price movements exceed a predefined threshold, thereby avoiding cognitive bias Kaminski and Lo (2014). It automatically restricts trading before investors fully grasp the new price information or the underlying market shift Tschoegl (1988). However, if not properly managed, stop-loss strategies may trigger feedback effects—such as price cascades—by amplifying losses in high-noise environments (Osler 2003). Hence, it is essential to incorporate volatility filtering into a stop-loss mechanism. Rather than using a fixed threshold, volatility filtering adjusts the stop level based on recent market volatility, ensuring that the strategy is not whipsawed by transient volatility spikes.

Dai et al. (2021) argue that trailing stop-loss strategies can reduce overall risk and limit downside exposure during market declines. Thus, incorporating stop-loss mechanisms reflects real-world trading constraints and risk management practices. This implementation, with its dynamic adjustments based on the spread's rolling volatility, offers a more sophisticated risk control measure than static thresholds. James and Yang (2010) further point out that a well-calibrated stop-loss tool acts as an effective failsafe in risk management.

This study selects the ten largest cryptocurrencies by market capitalization—equally divided between proof-of-work (PoW) and proof-of-stake (PoS) consensus mechanisms—and examines all 90 unique pairwise permutations for evidence of cointegration. The analysis has two primary objectives: first, to evaluate whether portfolios composed exclusively of PoW or PoS assets deliver superior risk-adjusted returns; and second, to determine whether combining consensus mechanisms increases the prevalence of stable cointegrated relationships. Adelopo and Luo (2025) provide empirical evidence that the consensus mechanism is a key determinant of hedging effectiveness and asset comovement. Milunovich (2022) documents that PoW coins exhibit stronger network linkages and greater spillover effects than PoS coins. Huang et al. (2023) demonstrate that portfolios diversified across consensus categories—combining both PoW and PoS assets—deliver the greatest diversification benefits. By grouping cryptocurrencies into homogeneous classes, aggregated portfolio returns tend to exhibit distributions that more closely approximate a normal distribution than those of individual assets, consistent with the central limit theorem.

The empirical analysis encompasses several robustness tests to evaluate strategy performance and validity (in-sample and out-of-sample analysis). I examine the sensitivity of results to various training/testing split ratios and employ walk-forward validation. This validation methodology systematically divides the data set into multiple overlapping training and testing periods, enabling dynamic strategy recalibration within each testing window. This approach effectively simulates real-time decision-making under evolving market conditions, enhancing the strategy's robustness and practical applicability. In addition, I benchmark the strategy's performance against momentum-based approaches to provide additional context for its effectiveness.

The empirical results demonstrate that the proposed pairs trading strategy exhibits superior performance not only compared to trivial pairs trading stragegy but also with the buy-and-hold approach, as evidenced by higher returns, Sharpe ratios and reduced drawdown metrics. The analysis highlights the significant impact of the lookback period on strategy returns. This sensitivity becomes particularly apparent in walk-forward validation, where certain periods yield negative returns. To investigate this phenomenon further, I conduct a comparative analysis between fixed and dynamically optimized lookback periods. The results reveal that minor variations in the lookback parameter can produce substantial performance differentials, emphasizing the importance of careful parameter selection and validation for strategy robustness. Thus, to address the critical issue of parameter overfitting in trading strategies, this paper introduces a parameter degradation analysis framework designed to evaluate how strategy performance deteriorates when transitioning from in-sample (IS) calibration to multiple consecutive out-of-sample (OOS) periods. Unlike conventional validation methods, which typically rely on single train-test splits, this framework tracks parameter robustness across sequential forward-looking windows. Consequently, it enables the identification of optimal parameter ranges and effectively distinguishes robust from spurious IS performance. I demonstrate the effectiveness of this framework by applying it to cryptocurrency pairs trading strategies. Through this analysis, I compare different optimization techniques—including nested cross-validation and genetic algorithms—to assess their varying degrees of susceptibility to overfitting and their ability to generalize under changing market conditions. This evaluation provides robust validation for parameter selection before implementing trading features, such as volatility filtering and trailing stop-loss mechanisms.

This paper makes five novel contributions to the empirical finance literature on cryptocurrency trading strategies. First, it demonstrates that our optimized pairs-trading framework systematically outperforms both a conventional pairs-trading benchmark and passive (buy-and-hold) approaches across multiple risk-adjusted performance metrics—including Bitcoin, which delivered the highest standalone return over the sample period. Second, to our knowledge, this is the first study to integrate Zegadlo (2022) bull–bear regime-dating algorithm into a cryptocurrency pairs-trading context. Despite the use of a predominantly bullish sample, the optimized strategy produces positive returns in both bull and bear regimes and outperforms passive cryptoasset strategies during bullish phases. This challenges Chu et al. (2020)'s finding that buy-and-hold dominates active approaches in rising markets. Third, incorporating a dynamic volatility filter together with a trailing stop-loss mechanism materially reduces volatility and maximum drawdowns relative to a baseline pairs-trading implementation, underscoring its effectiveness in controlling downside risk. Fourth, the results reveal that the consensus mechanism might influence cointegration frequency and profitability: PoS pairings exhibit the highest incidence of cointegration, whereas PoW pairs—despite having the lowest cointegration rate—achieve the strongest returns. Finally, the study highlights the critical importance of calibrating the lookback period for strategy performance, providing actionable guidance for parameter optimization in systematic trading system design.

This paper is structured as follows: Section 2 provides an overview of the data, including the selection process and key characteristics. Section 3 outlines the methodology employed in the analysis. Section 4 presents the implementation of the methodology and discusses the empirical results. Finally, Section 5 concludes the study and offers suggestions for future research.

2 | Data

A sample of cryptocurrencies is selected based on market capitalization and consensus mechanisms. In blockchain technology, transactions are validated and network security is ensured through two primary algorithms. The first is Proof of Work (PoW), a consensus algorithm where miners compete to solve complex mathematical problems using computational power. The miner who successfully solves the problem is rewarded with cryptocurrency. The second method is Proof of Stake (PoS), where transaction validation is based on the number of coins held by users who are willing to lock them ('stake') as collateral for network security. Each approach has its drawbacks: PoW is criticized for its high energy consumption and reliance on specialized hardware, which raises environmental concerns. PoS, on the other hand, faces criticism for potentially centralizing power among users with larger coin holdings, giving them greater influence over transaction validation.

This study analyzes the five largest cryptocurrencies by market capitalization, categorized by their consensus mechanism—proof-of-work (PoW) or proof-of-stake (PoS). Although Ripple (XRP) uses the Ripple Protocol Consensus Algorithm (RPCA), it is classified with PoS cryptocurrencies due to its comparable energy efficiency and transaction validation methodology. The analysis also accounts for Ethereum's transition from PoW to PoS in September 2022, a pivotal event in cryptocurrency evolution. As Liu et al. (2025) demonstrate, the PoS consensus offers significant advantages over PoW, including substantially lower energy consumption, greater market liquidity, enhanced operational efficiency, and reduced price volatility.

This selection yields ten cryptocurrencies, resulting in 90 potential pairs for analysis. To determine each pair's suitability for the proposed trading strategy during the IS period, I employ Engle and Granger (1987)'s cointegration approach. This test reveals that 37 of the 90 potential pairs exhibit cointegration, forming the basis for the subsequent analysis. The data set comprises daily price data for these cryptocurrencies from January 1, 2019, to May 29, 2024, denominated in US dollars and sourced from CoinMarketCap. Table 1 presents an overview of the sample, detailing the symbol, full name, and consensus mechanism for each cryptocurrency included in the study.

Table 2 shows the descriptive statistics in level and log-returns. We reject the null-hypothesis of unit root for the log-returns using the Augmented Dickey-Fuller (Dickey and Fuller 1979) and

Phillip-Peron tests (Phillips and Perron 1988). Figure 1 depicts the Pearson correlation with the involved variables, being the pairs ETH and Binance Coin and Ethereum classic and Dogecoin reaching the highest correlation (95%) of all pairs. Figure 2 illustrates the time series of the cryptocurrencies in our sample.

TABLE 1 | Cryptocurrency sample characteristics.

Symbol	Name	Consensus mechanism
ADA. USD	Cardano	PoS
BNB. USD	Binance coin	PoS
ETH. USD	Ethereum	PoS
XRP. USD	Ripple	Neither (RPCA)
TRX. USD	TRON	PoS
BCH. USD	Bitcoin cash	PoW
BTC. USD	Bitcoin	PoW
DOGE. USD	Dogecoin	PoW
ETC. USD	Ethereum classic	PoW
LTC. USD	Litecoin	PoW

3 | Methodology

I present a cointegration-based pair trading (active) strategy that adapts dynamically to market conditions by leveraging statistical signals—the Z-score—of asset price spreads (Palomar 2025). The first step is to find the cointegrated pairs. Let $P_{1,t}$ and $P_{2,t}$ represent the prices of two assets at time t. $Y_{i,t} = \log(P_{i,t})$, represents the log-transformed prices. I assume the existence of a cointegrating relationship of the following form:

$$Y_{1,t} = \mu + \gamma Y_{2,t} + \epsilon_t, \quad \epsilon_t \sim I(0), \tag{1}$$

where μ is the intercept, γ is the cointegrating coefficient, and ϵ_t is a stationary error term. This implies that the spread S_t:

$$\text{Spread } (S_t) = Y_{1,t} - \gamma Y_{2,t} \tag{2}$$

is mean-reverting, and thus can be used for a trading strategy based on statistical arbitrage.

To generate trading signals, I normalize the spread by calculating the Z-score, which measures the spread's deviation from its rolling mean, scaled by its rolling standard deviation. The Z-score is defined as:

TABLE 2 | Descriptive statistics.

Symbol	ADF test	PP test	Mean	Std	Min	Max	Kurtosis	Skewness
Levels								
ADA. USD	0.67	0.83	0.53	0.59	0.02	2.97	2.46	1.68
BNB. USD	0.55	0.62	222.00	185.00	5.57	676.00	−0.90	0.37
ETH. USD	0.65	0.68	1,534.00	1,211.00	105.00	4,812.00	−0.72	0.50
XRP. USD	0.31	0.09	0.50	0.28	0.14	1.84	2.51	1.50
TRX. USD	0.12	0.11	0.06	0.03	0.01	0.17	−0.72	0.43
BCH. USD	0.22	0.09	310.00	189.00	89.40	1,542.00	5.71	1.82
BTC. USD	0.80	0.82	27,150.00	18,073.00	3,399.00	73,084.00	−0.68	0.59
DOGE. USD	0.37	0.18	0.08	0.09	0.00	0.69	4.94	1.85
ETC. USD	0.15	0.06	20.70	17.30	3.47	134.00	4.17	1.68
LTC. USD	0.32	0.27	93.50	53.20	30.30	386.00	3.63	1.75
Log-returns								
ADA. USD	***	***	0.00	0.05	−0.50	0.28	7.46	−0.20
BNB. USD	***	***	0.00	0.05	−0.54	0.53	22.20	−0.21
ETH. USD	***	***	0.00	0.04	−0.55	0.23	15.40	−1.19
XRP. USD	***	***	0.00	0.05	−0.55	0.55	22.10	0.47
TRX. USD	***	***	0.00	0.05	−0.52	0.33	15.00	−0.84
BCH. USD	***	***	0.00	0.05	−0.56	0.46	16.60	0.04
BTC. USD	***	***	0.00	0.04	−0.47	0.17	18.30	−1.17
DOGE. USD	***	***	0.00	0.07	−0.52	1.52	120.00	5.99
ETC. USD	***	***	0.00	0.06	−0.51	0.35	10.90	0.13
LTC. USD	***	***	0.00	0.05	−0.45	0.27	9.92	−0.74

Note: This table provides descriptive statistics for levels and log-returns of cryptocurrency prices. The ADF (Augmented Dickey-Fuller, Dickey1979) and PP (Phillips-Perron, Phillips1988) tests assess the presence of a unit root. Mean, standard deviation (Std), minimum (Min), and maximum (Max) values are included. Kurtosis and skewness summarize the distributions. Statistical significance of the ADF and PP tests is denoted by *** (1% level), ** (5% level), and * (10% level).

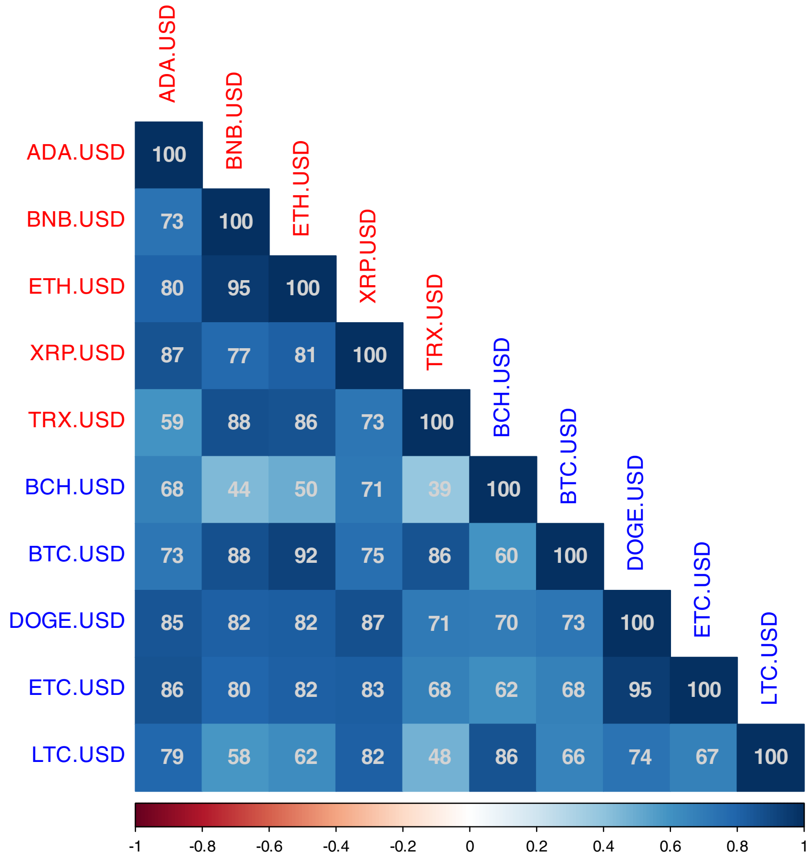

FIGURE 1 | Pearson correlation of the log-returns of the cryptocurrencies. *Source:* Coinmarketcap.

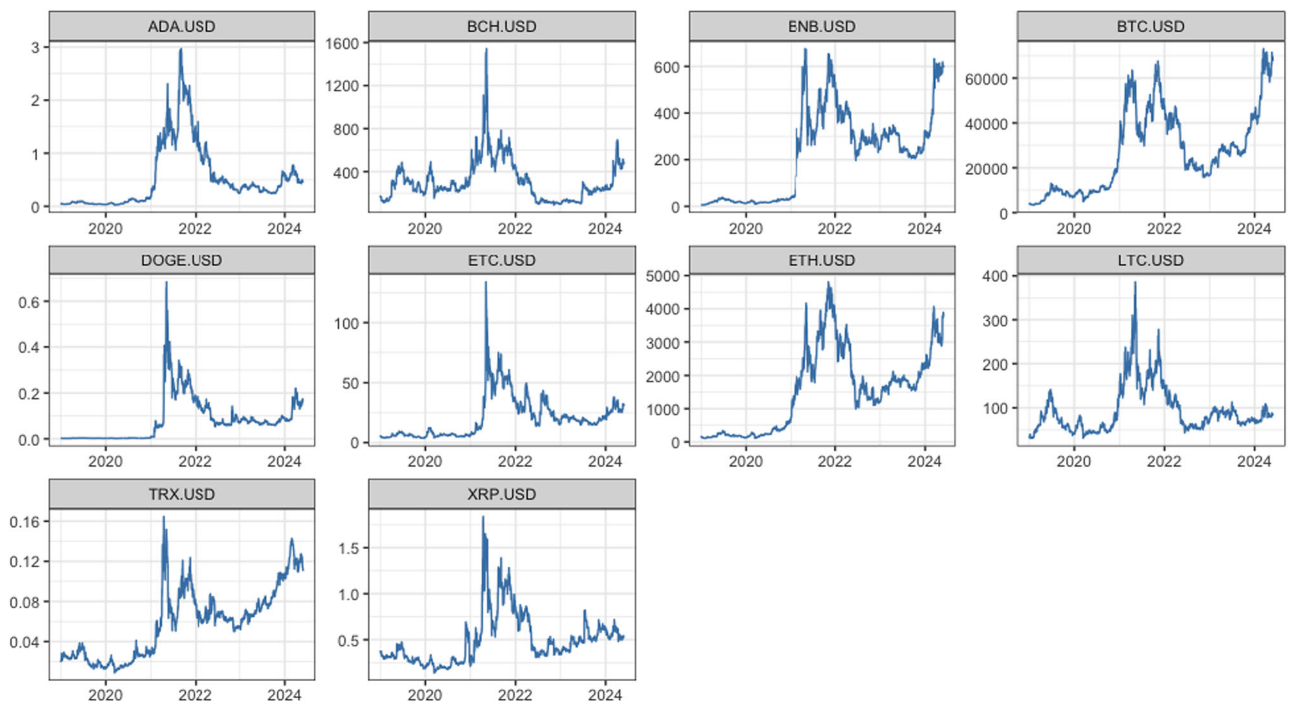

FIGURE 2 | Cryptocurrencies time series in level. *Source:* Coinmarketcap.

$$Z_t = \frac{S_t - \mu_{S_t}}{\sigma_{S_t}} \tag{3}$$

where μ_{S_t} is the spread's rolling mean over a lookback period L, and σ_{S_t} is the rolling standard deviation of the spread over the same period.

The threshold values ($Threshold_{long}$ and $Threshold_{short}$) are set symmetrically at ± 0.7 standard deviations of the spread. The threshold parameter is applied to the standardized spread (Z-score) to generate trading signals. Thus, trading signals are generated based on thresholds determined by the Z-score:

$$S_t = \begin{cases} 1 & \text{if } Z_t \le \text{Threshold}_{\text{Long}} \\ -1 & \text{if } Z_t \ge \text{Threshold}_{\text{Short}}, \\ 0 & \text{otherwise} \end{cases}$$

where S_t represents the position taken at time t (1 for long, −1 for short, and 0 for no position). $\text{Threshold}_{\text{long}}$ and $\text{Threshold}_{\text{short}}$—are dynamically adjusted based on rolling volatility.

I account for transaction cost, where c define the transaction cost as a percentage of the trade amount, and R_t is the realized return after accounting for the c:

$$R_t = \text{Signal}_{t-1} \cdot \Delta S_t - c \cdot \mid \Delta\text{Signal}_t \mid \tag{4}$$

where $\Delta S_t = S_t - S_{t-1}$ is the change in the spread, and $|\Delta\text{Signal}_t|$ accounts for the cost when positions are opened or closed.

To avoid extreme market conditions, particularly in the cryptocurrency market, I implement a volatility filtering mechanism to reduce risk exposure during periods of severe market turbulence. The filter is developed through a multi-step process based on spread returns.

First, I calculate the rolling volatility of spread returns using a specified lookback window L_v:

$$\sigma_t = \sqrt{\frac{1}{L_v}\sum_{i=t-L_v+1}^{t}(r_i - \bar{r})^2} \tag{5}$$

where r_i represents the spread returns and $\bar{r}$ is the mean return over the lookback period. The spread volatility is calculated on a rolling basis with a 30-day window ($Vol_{lookback} = 30$). The average volatility level $\bar{\sigma}$ is then computed across the entire sample period:

$$\bar{\sigma} = \frac{1}{T}\sum_{t=1}^{T}\sigma_t \tag{6}$$

The volatility filter is defined as a binary indicator function:

$$\text{VolFilter}_t = \begin{cases} 1 & \text{if } \sigma_t \le k\,\bar{\sigma} \\ 0 & \text{if } \sigma_t > k\,\bar{\sigma} \end{cases} \tag{7}$$

where k represents the volatility threshold multiplier, set to 1.5.

This filter is then applied to modify the trading signals:

$$\text{Signal}_t^{\text{filtered}} = \text{Signal}_t \cdot \text{VolFilter}_t \tag{8}$$

The volatility filter is designed to suspend trading activity during periods of heightened market volatility, specifically when the current volatility surpasses 1.5 times the average volatility level. This feature acts as a risk management measure, mitigating exposure to turbulent market conditions where the statistical relationships underpinning the strategy may become less reliable.

The performance of the volatility filter is influenced by two key parameters: the volatility estimation window L_v and the threshold multiplier k. The estimation window L_v determines how quickly the volatility measure adapts to market fluctuations, while the multiplier k dictates the strictness of the volatility threshold. These parameters can be fine-tuned to align with the characteristics of the assets being traded and prevailing market conditions.

In addition to the volatility filter, the strategy incorporates a dynamic trailing stop-loss mechanism to limit trading losses. This mechanism adjusts stop-loss levels dynamically in response to both price movements and prevailing market volatility, ensuring that the risk management process remains adaptive to changing market conditions. For each position, the mechanism tracks price extremes (highest H_t for long positions, lowest L_t for short positions) that occur after entering the trade. These levels are then used to determine stop-loss triggers. So, for a long position opened at time t entry

$$H_t = \max_{s \le t}\{S_s : s \ge t_{\text{entry}}\} \tag{9}$$

where, t_{entry} is the time when the position was initiated, S_s represents the spread value at time s, and H_t is the highest spread value achieved up to time t. The position is closed if the spread falls below $H_t(1 - d)$, where d is the stop-loss threshold ($d = 2.5\%$). Similarly, for a short position:

$$L_t = \min_{s \le t}\{S_s : s \ge t_{\text{entry}}\} \tag{10}$$

L_t is the lowest spread achieved up to time t. In this case, the position is closed if the spread rises above $L_t(1 + d)$.

The dynamic stop-loss threshold incorporates current market volatility through a scaling factor:

$$\lambda_t = \lambda \cdot \max\left(\frac{\sigma_t}{\bar{\sigma}}, 1\right) \tag{11}$$

where λ is the base stop-loss factor (default: 0.025), σ_t is the current volatility, and $\bar{\sigma}$ is the average volatility

I split the data into in- and out-of-sample periods in a 75%–25% proportion. The cointegrating vector $(\hat{\mu}, \hat{\gamma})$ is estimated using the IS data. The Z-scores and trading signals are generated for the IS period. In the OOS period, the parameters $\hat{\mu}$ and $\hat{\gamma}$ are kept fixed, and trading signals are generated using the same method as in the IS period.

The lookback period L is a critical parameter that directly impacts the calculations of rolling mean, standard deviation, and Z-score. To determine the optimal lookback period, the strategy performs a single-step grid search over a predefined range $\mathscr{L}$, incremented by 1 day. The objective is to identify the lookback period that yields the highest Sharpe ratio during the IS period:

$$L^* = \arg\max_{L \in \mathscr{L}}(\text{Sharpe Ratio}_{\text{in-sample}}(L)). \tag{12}$$

The optimal lookback period L^* is then used to generate trading signals and evaluate the strategy's performance during the OOS period. The strategy's performance is evaluated using the following metrics:

1. Cumulative return: $\prod_{t=1}^{T}(1 + R_t) - 1$

2. Sharpe ratio: $\frac{\mathbb{E}[R_t] - R_f}{\sigma_{R_t}}$ ($R_f = 4\%$ annual)

3. Maximum drawdown (MDD): $\max_t (\max_{\tau \le t} X_\tau - X_t)$

4. Sortino ratio: $\frac{\mathbb{E}[R_t - R_f]}{\sigma_d}$. σ_d is the standard deviation of negative returns.

4 | Empirical Results

This section empirically implements and tests the pairs trading strategy. The process begins by running the strategy, as described in the Methodology section, for all cointegrated pairs. Among the tested pairs, the pair with the highest Sharpe ratio in the sample is selected, with LTC and DOGE emerging as the top-performing pair.[1] Figure 3 illustrates the trading signals generated alongside the Z-score, identifying entry and exit points for both long and short positions. Additionally, the cumulative return of the strategy is depicted, $R_t = \prod_{i=1}^{t}(1 + r_i) - 1$, where R_t represents the cumulative return up to time t, and r_i represents the return at time i. The data set is divided into a training set (75%) and a testing set (25%), with the optimal lookback period identified using the training data and subsequently applied to evaluate performance in the testing period.

The results presented in Table 3 compare the performance of traditional pairs trading and buy-and-hold strategies for LTC, DOGE, and BTC against the optimized pairs-trading approach developed in this study. BTC is included because it exhibits the highest return among the cryptocurrencies in the sample. Simple pairs trading involves implementing an arbitrage strategy based on the mean reversion of price spreads, following a

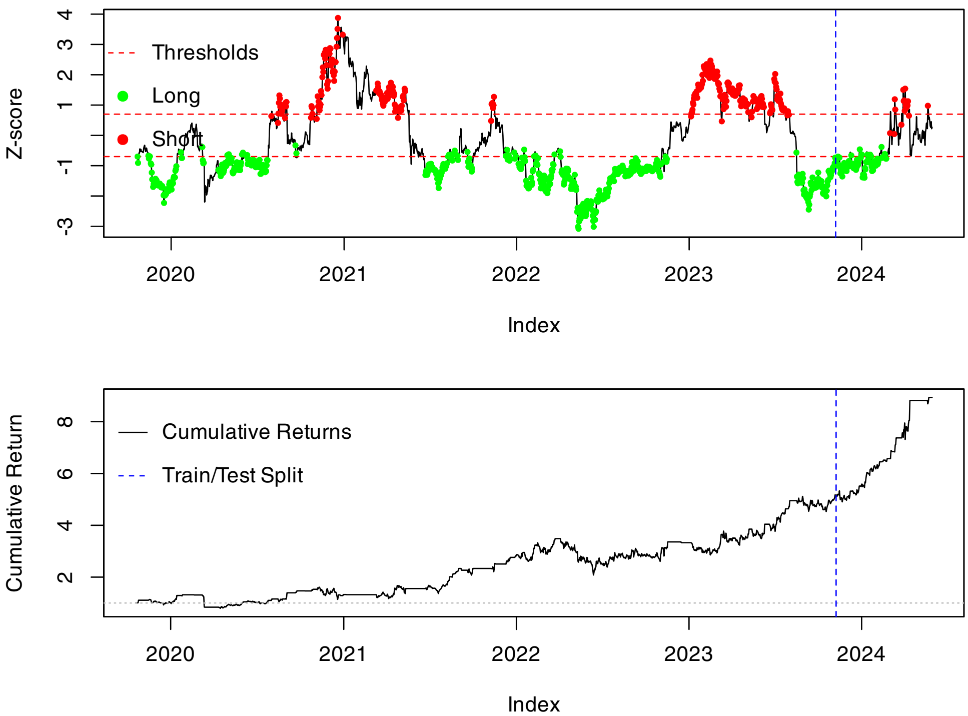

FIGURE 3 | The upper plot displays the Z-score time series as a black line, with the upper and lower threshold lines indicated by red dashed lines. Green and red points represent the long and short trading signals, respectively, plotted at the corresponding Z-score values and time indices. The vertical blue dashed line denotes the split between the training and testing periods (75%–25% ratio). The lower plot presents the cumulative returns of the trading strategy over time, with the same train/test split indicated by the vertical blue dashed line. The horizontal gray dashed line at $y = 1$ represents the starting point of the cumulative returns.

TABLE 3 | Comparison of trading strategy performance.

Strategy	Return (Ann.)	Sharpe ratio	Sortino ratio	Calmar ratio	MDD	Volatility
Pairs Trading (LS)	0.22	0.54	0.06	0.96	0.23	0.31
Pairs Trading (Optimized)	0.71	2.12	0.19	4.95	0.14	0.30
LTC Buy & Hold	−0.17	−0.36	−0.01	−0.34	0.51	0.57
DOGE Buy & Hold	0.13	0.12	0.04	0.27	0.47	0.66
BTC Buy & Hold	0.61	1.37	0.13	2.70	0.23	0.40

two-step procedure (Chan 2013; Palomar 2025): (1) estimating the hedge ratio during the training phase and (2) generating trading signals based on standard spread deviations. The optimized pairs-trading strategy is found to outperform both simple pairs trading and the passive buy-and-hold approach across all individual cryptocurrencies and performance metrics (annual return, Sharpe ratio, Sortino ratio, Calmar ratio, MDD, and volatility). Figure 4 illustrates the OOS cumulative returns for both pairs-trading strategies, as well as the buy-and-hold strategies, for LTC, DOGE, and BTC. The optimized pairs-trading strategy exhibits a steady increase in cumulative returns with lower variability than the other strategies, consistent with the performance metrics in Table 3. Moreover, the Calmar ratio (Calmar Ratio $= \frac{\text{AnnualizedReturn}}{\text{MaximumDrawdown}}$) highlights the superior risk-adjusted return for the pairs-trading strategy. In particular, the strategy's Calmar ratio is notably higher than that of all other strategies, demonstrating not only its effectiveness in managing risk and delivering consistent returns but also the benefit of incorporating a dynamic stop-loss mechanism.

Next, I evaluate strategy performance by grouping cointegrated pairs according to their consensus-mechanism combinations to assess potential diversification benefits. Table 4 the prevalence of cointegration across the 90 possible pairwise combinations, of which 37 exhibited statistically significant cointegration. PoS–PoS pairs displayed the highest cointegration rate (70%), followed by PoW–PoS (44%) and PoS–PoW (36%), whereas PoW–PoW pairs had the lowest rate (15%). These results imply that the PoS consensus may foster more persistent equilibrium relationships between asset prices. However, Table 5 reveals an inverse relationship between cointegration prevalence and trading performance. Despite their low cointegration incidence, PoW–PoW pairs achieved the highest average Sharpe ratio (0.65), while all other consensus combinations produced negative Sharpe ratios. Moreover, PoW–PoW pairs incurred the fewest stop-loss activations (80), compared with 123 for PoW–PoS, 152 for PoS–PoW, and 135 for PoS–PoS, indicating more stable trading outcomes under a PoW–PoW regime.

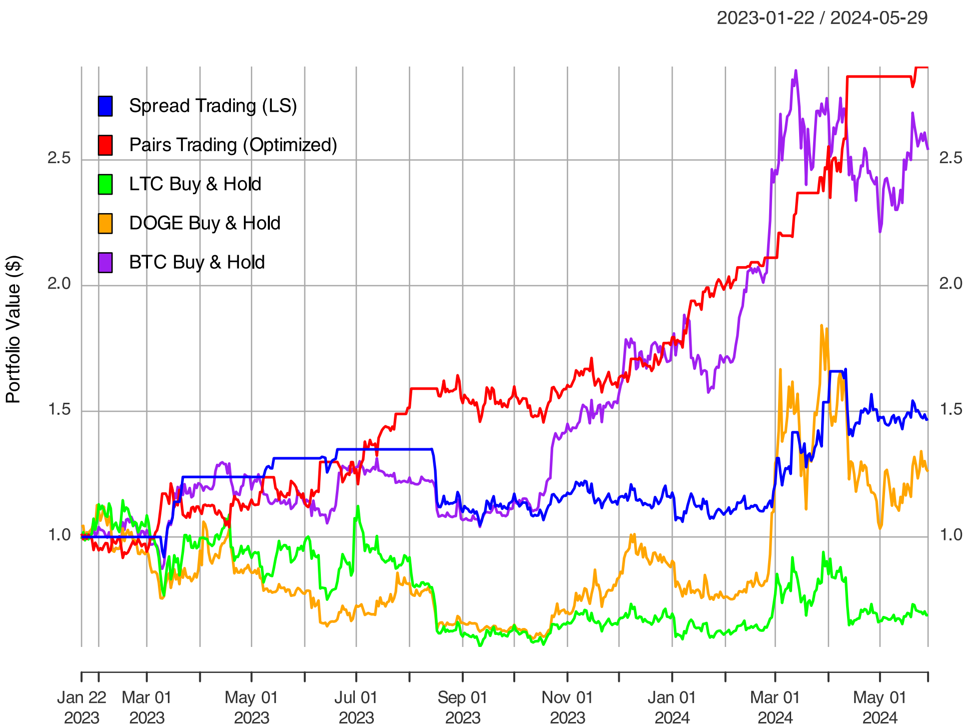

FIGURE 4 | Comparison of out-of-sample cumulative returns for trading strategies: Spread Trading (LS) (blue), Pairs Trading (Stop Loss) (red), LTC Buy & Hold (green), DOGE Buy & Hold (orange), and BTC Buy & Hold (purple). The initial portfolio value starts at $1.

TABLE 4 | Cointegration frequency by consensus mechanism.

Consensus pair	Total pairs	Cointegrated pairs	% Cointegrated (group)
PoW-PoW	20	3	15% (3/20)
PoW-PoS	25	11	44% (11/25)
PoS-PoW	25	9	36% (9/25)
PoS-PoS	20	14	70% (14/20)
Total	90	37	41.1% (37/90)

TABLE 5 | Performance metrics for cointegrated pairs by consensus mechanism.

Consensus	Count	Sharpe ratio	Cum. return	Stop-Loss hits	Lookback (days)
PoW–PoW	3	**0.54**	**0.65**	80.3	331
PoW–PoS	10	−0.98	−0.52	122.6	296
PoS–PoW	10	−1.42	−0.50	151.7	260
PoS–PoS	14	−1.23	−0.86	134.6	218
Total	37	100%			

In Appendix C, I incorporate one additional PoW cryptocurrency (Monero) and two less liquid PoS cryptocurrencies (Cronos and Fantom), while excluding XRP-USD due to its use of the RPCA, which differs from both PoW and PoS.[2] Subsequently, I re-ran the analysis, resulting in a total of 132 permutations. Notably, despite the inclusion of these three additional cryptocurrencies, the results remained consistent with those reported in Tables 4 and 5. Figure C1 illustrates key differences across categories: PoW–PoW pairs display the highest average Sharpe ratios and cumulative returns, whereas PoS–PoW pairs contain the highest number of cointegrated pairs but yield the lowest overall performance. Figure C2 indicates that PoW–PoW pairs experience the largest average maximum drawdowns but exhibit fewer stop-loss hits compared to other consensus pair categories. Additionally, I examine the statistical significance of differences across consensus mechanisms and cointegration status in Table C1. The global ANOVA tests do not reveal statistically significant differences in Sharpe ratios among the consensus categories. Pairwise comparisons suggest only a marginal advantage of PoW–PoW pairs over PoS–PoS pairs at roughly the 10% significance level. Therefore, the results imply that the consensus mechanism alone does not fully account for variations in risk-adjusted returns within this analysis. Any observed advantage for PoW–PoW pairs should thus be interpreted with caution.

4.1 | Identifying Bull and Bear Regimes

In a bull market, a buy-and-hold strategy might appear appealing, but accurately timing market peaks—and deciding whether the market is in a bullish or bearish phase—remains challenging. Pagan and Sossounov (2003) identify distinct phases of the asset-price cycle, marked by statistically detected turning points, and create a rules-based algorithm to find local peaks (bull-to-bear transitions) and troughs (bear-to-bull transitions) directly from price data. They demonstrate that the duration and amplitude of these phases are endogenous to the underlying data-generating process, suggesting that even a simple random-walk model can reproduce empirically realistic bull and bear markets. Zegadlo (2022), in turn, proposes a single discrete parameter–adjusted to the data—s periodicity–that removes subjectivity from the regime identification process.

Thus, I implement algorithm proposed by Zegadlo (2022) (hereinafter "PZ") to identify bull and bear market phases and compare them with the optimized pairs-trading strategy. This approach provides a systematic framework for assessing the strategy's performance across distinct market regimes. The PZ algorithm first detects all local maxima and minima in the price series, then iteratively filters these candidate turning points by retaining only those that are true extrema relative to their immediate neighbors, thereby enforcing an alternating sequence of peaks and troughs. Once the final set of turning points is established, the intervals between successive peaks and troughs are classified as either bull or bear regimes.

I begin by analyzing market regimes during the IS period for each cryptocurrency using the PZ algorithm. Figure 5 reveals that half of the cryptocurrencies spent over 60% of their time in bull regimes, with only three cryptocurrencies remaining in bull markets for less than 50% of the period. These findings indicate that the cryptocurrency market was predominantly bullish during the IS analysis period.

To gain a broader perspective of the market regime, I construct a portfolio weighted by the liquidity of all cryptocurrencies in the sample (measured by turnover—trading volume multiplied by the respective cryptocurrency price). Figure 6 illustrates the regime phases across the entire analyzed period. Applying the PZ algorithm reveals a regime distribution of ~62.5% bull market days and 37.5% bear market days, confirming the predominantly bullish nature of the market. Notably, both regimes yielded positive returns; however, the cumulative return was substantially higher during bull market phases.

Table 6 presents the strategy's performance across different market regimes. The OOS results demonstrate noteworthy performance when compared with the IS results. While bull markets predominated during the OOS period, the strategy achieved an exceptional Sharpe ratio (2.24) during bear market conditions, suggesting robust adaptability. The profit factor ($\frac{\text{TotalProfitfromWinners}}{\text{TotalLossfromLosers}}$) remained consistently positive (>1) across all regimes, despite a win rate of ~32%. The volatility filtering and trailing stop-loss mechanisms likely contribute to this favorable risk-reward profile by minimizing losses during drawdown periods.

To statistically evaluate the strategy's adaptability across market conditions, I conduct regime-dependent performance tests using bootstrap-based Sharpe ratio comparisons. I employ a stationary bootstrap procedure (Politis and Romano 1994) with 5000 replications to construct confidence intervals and test the significance of pairwise differences between regimes. This approach accounts for potential serial dependence while preserving the distributional properties of the returns series.

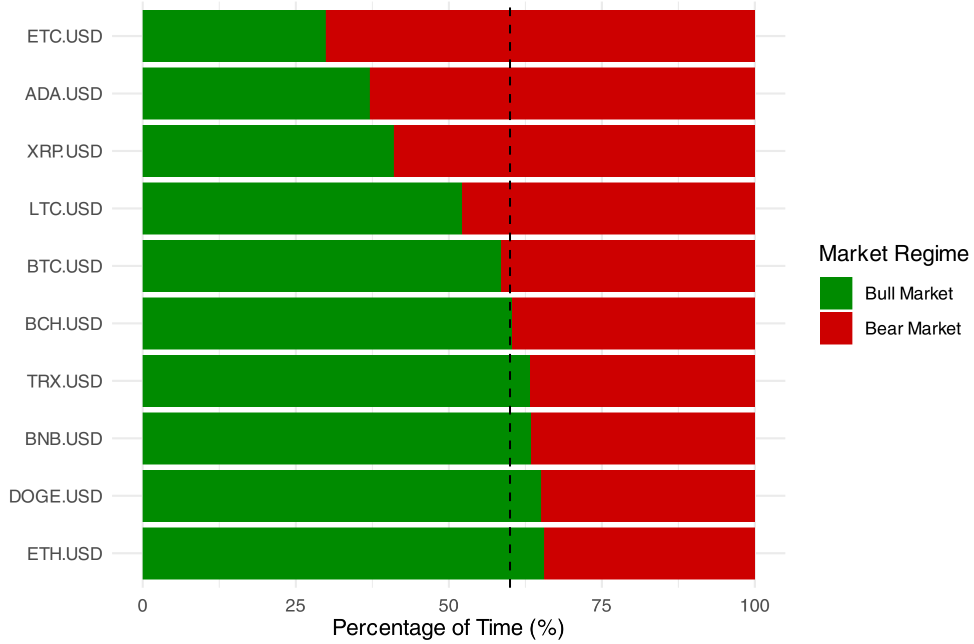

FIGURE 5 | Percentage of time spent in bull versus bear market regimes during the training period (January 2019–January 2023) for each cryptocurrency in the sample. The dashed horizontal line indicates the 60% threshold for regime dominance.

FIGURE 6 | The plot depicts the cumulative returns of the LTC-DOGE pairs trading strategy with line color indicating market regime (green for bull markets, red for bear markets). The vertical dashed line represents the division between the IS and OOS periods.

The statistical tests (Table 7) reveal that while some differences are economically substantial—particularly the 1.99 improvement in bear market Sharpe ratio—none achieve statistical significance at conventional levels. All comparisons yielded p-values exceeding 0.50. The limited statistical significance is likely attributable to the restricted sample size in the OOS bear regime (102 observations) and the inherent volatility of cryptocurrency returns. Nevertheless, the consistent outperformance across different market conditions provides practical evidence of the strategy's adaptability to changing market regimes.

4.2 | Robustness Checks

To ensure the optimized pairs trading strategy is robust and capable of outperforming under varied market conditions, I conduct four validation tests. First, a split-sample ratio test evaluates performance across multiple train–test splits. Second, walk-forward validation assesses the strategy over a series of rolling windows, simulating real-time re-estimation and deployment. Third, a sensitivity analysis of the lookback period—spanning 5 to 360 days—quantifies how different historical window lengths affect returns and risk metrics. Finally, I benchmark the pairs trading strategy against both a momentum-based strategy and a buy-and-hold approach to contextualize its relative performance.

4.2.1 | Split Ratios

In this section, I analyze the performance of the pairs trading strategy across different training and testing sets. I vary the split ratio between training and testing from 60% to 90%, in increments of 5%. The objective of this analysis is to assess the stability of the strategy's parameters and the consistency of its performance across different training/testing splits. There are several advantages to testing different split ratios. First, it allows for evaluating the strategy across various market conditions, such as different trends or volatility regimes, ensuring that it does not rely on specific market environments to perform well. This contrasts with strategies such as momentum or passive investing, which often depend on trending markets to be effective. Second, testing different splits helps identify potential overfitting, where the model may perform well on a specific training set but fails to generalize to OOS data.

Table 8 presents the performance metrics for each split ratio, ordered by Sharpe ratio from highest to lowest. With a training set of 90% and a testing set of 10%, the strategy achieves an impressive Sharpe ratio of 3.97 and the smallest maximum drawdown of 7.94%. However, it is noteworthy that changes in the lookback period appear to significantly impact returns, emphasizing the importance of selecting this parameter carefully. In contrast, the number of stop-loss events remains relatively consistent across all split ratios.

TABLE 6 | Strategy performance by regime (Key metrics).

Regime	Obs.	Cum. return (%)	Sharpe	Max DD (%)	Win rate (%)	Profit F.
Market regimes						
Bull	1092	598.39	1.36	−29.36	31.59	1.38
Bear	593	27.93	0.47	−41.92	32.38	1.11
Out-of-Sample performance						
OOS Bull	392	127.30	1.84	−14.40	38.78	1.46
OOS Bear	102	26.26	2.24	−8.55	38.24	1.59
In-Sample						
IS Bull	700	207.26	1.17	−29.36	27.57	1.35
IS Bear	491	1.33	0.21	−40.75	31.16	1.05

TABLE 7 | Statistical tests of sharpe ratio differences between regimes.

Comparison	Difference	95% CI	*p* value	Sig.
Bull vs. Bear	0.89	[−0.68, 2.60]	0.5212	—
OOS Bull vs. OOS Bear	−0.38	[−3.66, 3.22]	0.8244	—
IS Bull vs. IS Bear	0.97	[−0.85, 2.92]	0.5354	—
Bull OOS vs. IS	0.64	[−1.38, 2.48]	0.5962	—
Bear OOS vs. IS	1.99	[−1.37, 5.23]	0.5086	—

Note: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

TABLE 8 | Sensitivity analysis of split ratios.

Split ratio (training)	Sharpe ratio	Calmar ratio	Max DD	Lookback period	Stop losses	N. trades
90%	3.97	14.05	7.94	256	53	232
80%	2.44	4.45	17.54	278	51	214
75%	2.12	4.95	14.40	291	51	208
85%	1.88	3.21	17.62	255	53	228
65%	1.28	1.46	31.07	180	55	241
60%	0.79	0.82	37.42	360	44	161
70%	0.72	0.88	31.46	176	56	233

4.2.2 | Walk-Forward Validation

To further examine the stability of the strategy under varying market conditions, I implement walk-forward validation (WAFO), which iteratively slides a window over the historical data, encompassing both training and testing sets. The WAFO is a function used for backtesting the implemented trading strategy. WAFO evaluates a trading strategy by training it during one period and testing it in the next, then advancing sequentially in time. Let $Y = \{y_t\}_{t=1}^{T}$ be a time series, where T represents the total number of observations. For each window i, the training window is defined as $Y_{\text{train},i} = \{y_t\}_{t=i}^{i+k-1}$, where k is the window size, and the testing window is $Y_{\text{test},i} = \{y_t\}_{t=i+k}^{i+k+s-1}$, where s is the step size. The starting index i increments by step size s for each iteration, such that $i \in \{1, 1+s, 1+2s, ...\}$ while $i+k+s-1 \leq T$.

The parameter estimation for each window is:

$$\hat{\theta}_i = \underset{\theta}{\operatorname{argmax}} \quad \mathscr{L}(\theta \mid Y_{\text{train},i}) \tag{13}$$

where $\hat{\theta}_i$ is the vector of estimated parameters, $\mathscr{L}$ is the likelihood function, and $Y_{\text{train},i}$ represents the training period data. One contribution of this paper is the optimization of the lookback period, which determines the number of past observations used to compute the main parameters of the trading strategy. In the crypto market, which is characterized by high volatility and swift regime changes, determining the optimal amount of historical data for training is crucial for strategy performance.

The historical data are segmented into windows of approximately 2 years, with a step size of 60 days, indicating how far the window shifts in each iteration. In each iteration, the strategy is trained on the window's training set, and the resulting returns and trading signals are recorded to estimate the performance metrics for the corresponding testing set. Unlike traditional cross-validation methods, this technique preserves the temporal order (Zhang and Zhang 2023). Similar to the split ratio analysis performed in the previous section, WAFO evaluates the strategy's adaptability to changing market dynamics. It offers a more realistic assessment of the strategy's performance by simulating real-time decision-making and continuously recalibrating the model to reflect evolving market conditions.

Table 9 presents the performance metrics for each step size. Notably, the Sharpe ratio remained positive for over 60% of the testing windows, indicating consistent risk-adjusted returns across varying market conditions. However, the substantial variations in the optimal lookback period merit particular attention. While the sensitivity of this parameter was initially observed during the split ratio analysis, the WAFO provides more definitive evidence of its significant impact on strategy returns.

The determination of an appropriate lookback period represents a fundamental trade-off between capturing long-term market trends and maintaining adaptability to short-term market dynamics. In such highly volatile and rapidly evolving markets, the quantity of historical data used to generate trading signals becomes a critical parameter for strategy optimization. A longer lookback period incorporates more extensive historical data, potentially capturing broader market trends and the mean-reverting behavior of the spread more accurately. However, extended lookback periods inherently reduce the strategy's sensitivity to short-term market fluctuations, potentially compromising its adaptability to regime shifts or sudden market variations. Conversely, shorter lookback periods may enhance responsiveness to recent market conditions but risk overreacting to transient noise, potentially increasing trading frequency and associated transaction costs.

4.2.3 | Out-Of-Sample and Lookback Period Sensibility Analysis

To examine the sensitivity of the lookback period on returns and drawdown, I modify the strategy to use fixed lookback periods instead of dynamically optimizing them. The range of lookback periods tested spans from five to 360 days. Figure 7 illustrates the relationship between the strategy's performance and the range of lookback periods. Returns exhibit significant variation with changes in the lookback period, with the highest returns achieved around 50–100 days. By contrast, the lowest drawdown occurs at a lookback period close to 350 days.

Moskowitz et al. (2012) analyze the significance of the time series momentum strategy by examining the *t-statistics* of the alpha across multiple lookback and holding periods. While the strategy consistently generates positive alphas, performance begins to decline as the lookback period extends beyond 12 months. By contrast, for the pairs trading strategy, no clear pattern emerges around the optimal lookback period, indicating a high sensitivity of returns to changes in this parameter. This sensitivity represents an important caveat, as it highlights the instability of the lookback period, which could limit the robustness of the pairs trading strategy.

Another important caveat is the risk of overfitting in forecasting models and trading strategies, a concern that has been examined in the existing literature. The familiar disclaimer “Past performance is no guarantee of future results” is widely used in financial markets to caution investors that historical success does not imply future returns. White (2000) highlights this risk in the context of forecasting models, emphasizing that seemingly strong results may arise by chance rather than reflecting true predictive merit. To address this concern, he proposes a method for detecting data snooping–the overfitting of models to historical data–by constructing bootstrap-based p-values for the null hypothesis that no strategy outperforms a given benchmark. Similarly, Bailey and de Prado (2014) warn against the manipulation of parameters to exclude specific trades that resulted in losses during the IS period. After a few iterations, a researcher may arrive at “optimal” parameters that exploit idiosyncratic features of the sample and are unlikely to generalize to the broader population. Furthermore, advances in computational power have made it easy to search through a vast number of potentially profitable trading strategies. The authors caution that this process often relies on pseudo-mathematical reasoning, spuriously validated through backtesting. Building on these insights, Bailey et al. (2016) define backtest overfitting

TABLE 9 | Walk-Forward validation and performance metrics.

Start	End	Lookback period	Sharpe ratio	Cum ret.	Max DD	N. trades	Stop loss
2023-02-18	2023-04-21	39	7.68	0.23	2.96%	20	10
2024-03-02	2024-05-03	258	6.20	0.27	2.97%	7	4
2022-08-13	2022-10-14	231	3.30	0.08	2.85%	10	8
2021-07-31	2021-10-01	299	2.62	0.20	10.23%	6	8
2021-10-02	2021-12-03	327	2.43	0.07	3.18%	3	7
2023-10-28	2023-12-29	141	2.00	0.07	7.85%	10	7
2021-12-04	2022-02-04	340	1.66	0.11	10.94%	13	5
2022-02-05	2022-04-08	302	1.55	0.12	13.89%	10	6
2021-05-29	2021-07-30	360	1.46	0.08	9.35%	5	7
2023-08-26	2023-10-27	92	1.37	0.05	6.32%	9	8
2020-07-18	2020-09-18	32	1.23	0.02	2.09%	6	3
2022-06-11	2022-08-12	27	1.22	0.11	15.74%	14	21
2023-06-24	2023-08-25	125	0.66	0.03	8.11%	4	9
2023-12-30	2024-03-01	321	0.59	0.03	6.61%	4	3
2020-11-21	2021-01-22	162	−0.49	−0.05	21.10%	1	4
2021-01-23	2021-03-26	303	−1.05	−0.02	4.22%	1	3
2022-04-09	2022-06-10	239	−1.29	−0.31	32.58%	10	12
2020-05-16	2020-07-17	83	−1.34	−0.27	28.24%	3	5
2021-03-27	2021-05-28	334	−1.44	−0.40	40.44%	4	9
2022-10-15	2022-12-16	344	−1.85	−0.21	22.94%	2	4
2020-09-19	2020-11-20	171	−1.91	−0.11	13.53%	5	3
2023-04-22	2023-06-23	330	−2.05	−0.08	9.48%	2	4
2022-12-17	2023-02-17	115	−2.10	−0.12	12.49%	3	7
Max.		360	7.68	0.27	40.44%	20	21
Min.		27	−2.10	−0.40	2.09%	1	3
Mean		216	0.89	0.00	12.53%	7	7
Std. Dev.		115	2.54	0.17	10.24%	5	4

as the scenario in which the best-performing IS strategy systematically underperforms the OOS strategy and identify this phenomenon as a primary reason why many algorithmic or systematic hedge funds fail to meet the elevated expectations raised by their managers.

Given the concerns raised by Bailey et al. (2016), Bailey and de Prado (2014), White (2000), I address these risks by analyzing Sharpe ratio degradation, parameter stability, and the overfitting ratio across a range of lookback periods. These approaches aim to identify robust parameter regions and isolate configurations likely to reflect spurious IS performance. I propose it as a practical and streamlined diagnostic toolkit for detecting the dynamics of overfitting.

To conduct an assessment of how perfomance decays across multiple OOS windows before implementing volatility filtering and trailing stop-loss mechanisms, consider two daily price series $\left\{P_t^{(1)}, P_t^{(2)}\right\}_{t=1}^{T}$, with T observations stored in Y. The sample is split chronologically at $\tau = \lfloor 0.5\,T \rfloor$. Observations $1 \le t \le \tau$ form the IS segment $\mathscr{D}_{\text{IS}}$, whereas the remainder, $t = \tau + 1, \ldots, T$, is partitioned into $K = \textit{forward\_windows}$ consecutive, nonoverlapping OOS windows $\{\mathscr{D}_{\text{OOS},k}\}_{k=1}^{K}$ of equal length $m = \lfloor (T - \tau)/\text{K} \rfloor$.

The only hyper-parameter varied systematically is the moving-average lookback length $\text{L} \in \mathcal{L} = \{30, 60, \ldots, 360\}$. The Z-score threshold is fixed at $\tau_z = \textit{threshold}_{value} = 1.5$. For each $L \in \mathscr{L}$ the hedge ratio is estimated on $\mathscr{D}_{\text{IS}}$ by the ordinary least-squares regression

$$P_t^{(1)} = \alpha_L + \beta_L P_t^{(2)} + \varepsilon_t, t = 1, \ldots, \tau \tag{14}$$

The resulting spread is $s_t = P_t^{(1)} - \beta_L P_t^{(2)} - \alpha_L$. Let $\mu_L^{\star} = \text{MA}_L(s_{\tau-L+1:\tau})$ and $\sigma_L^{\star} = \text{SD}_L(s_{\tau-L+1:\tau})$ denote, respectively, the moving-average and sample standard deviation of the most recent L IS spreads; both quantities are frozen for all

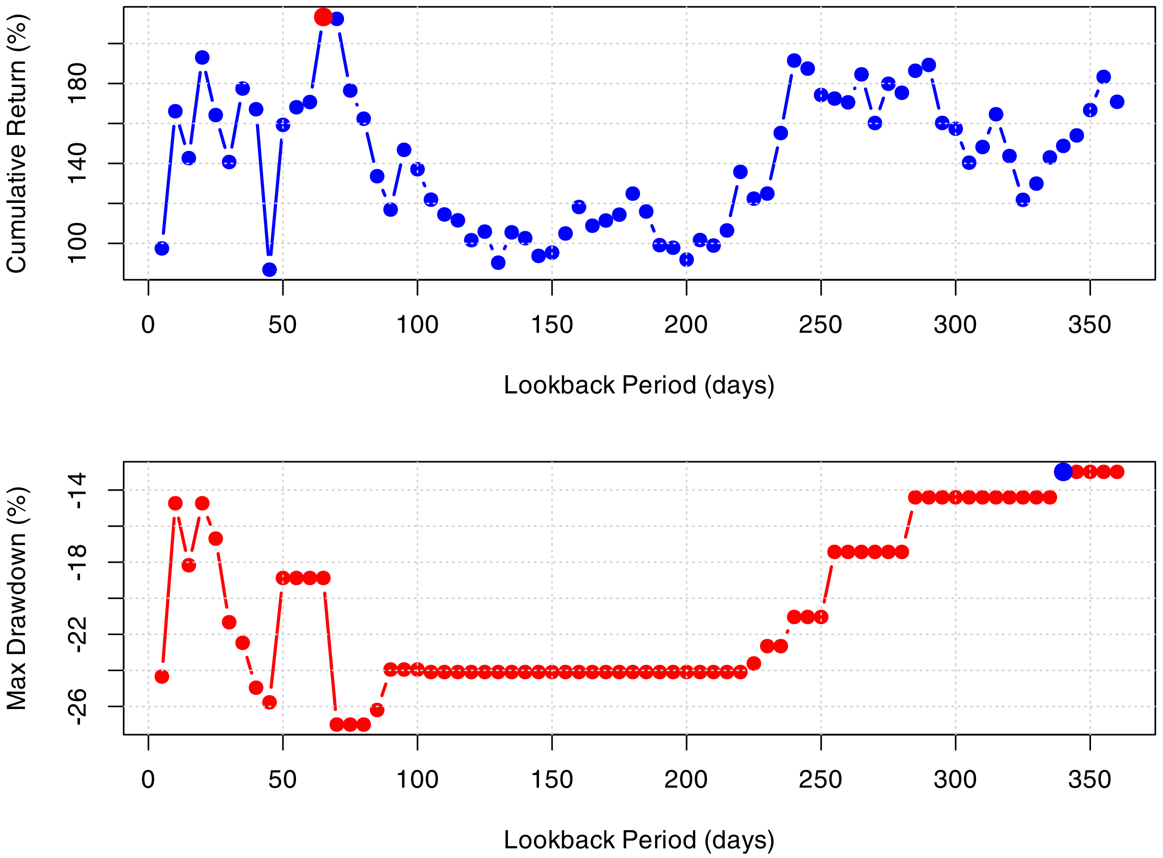

FIGURE 7 | The upper panel displays cumulative returns (%) across lookback periods, with the highest return highlighted in red. The lower panel shows maximum drawdowns (%), where the lowest drawdown is highlighted in blue.

subsequent evaluations. The IS Z-score is $z_t = \left(s_t - \mu_L^\star\right)/\sigma_L^\star$ and the trading signal is given by:

$$\text{signal}_t = \begin{cases} +1, & z_t \le -\tau_z, \\ -1, & z_t \ge \tau_z, \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

Daily strategy returns are computed as $r_t = \Delta s_t \text{signal}_{t-1}$, where $\Delta s_t = s_t - s_{t-1}$. Two performance indicators are retained: the annualized Sharpe ratio, $\hat{S}_{\text{IS}}(L)$, and the cumulative return, $C_{\text{IS}}(L) = \prod_{t=2}^{\tau}(1 + r_t) - 1$.

For every window $\mathscr{D}_{\text{OOS},k}$ $(k = 1, \ldots, K)$, the pre-estimated coefficients (α_L, β_L) and anchors $\left(\mu_L^\star, \sigma_L^\star\right)$ are held fixed. Spread, Z-score, signals and returns are re-computed exactly as in the IS block, yielding the pair of metrics $(\hat{S}_k(L), C_k(L))$.

The mean out-of-sample performance is defined as $\bar{P}_{\text{OOS}}(L) = K^{-1}\sum_{k=1}^{K} P_{\text{OOS},k}(L)$, where P denotes either Sharpe or cumulative return. The degradation quantity, $D(L) = P_{\text{IS}}(L) - \bar{P}_{\text{OOS}}(L)$, measures the loss of performance when moving from calibration to deployment. Stability across market regimes is quantified by:

$$S(L) = \left[\frac{1}{K}\sum_{k=1}^{K}\left(P_{\text{OOS},k}(L) - \bar{P}_{\text{OOS}}(L)\right)^2\right]^{1/2}, \tag{16}$$

and the over-fitting ratio is $R(L) = P_{\text{IS}}(L)/\bar{P}_{\text{OOS}}(L)$.

A parameter value L is regarded as robust when $D(L) \approx 0$, $R(L) \approx 1$, and $S(L)$ is small. Substantial positive degradation, $D(L) > 0$, or ratios significantly above unity indicate that the IS edge is unlikely to persist. Conversely, negative degradation may signal underfitting or structural change in the underlying spread dynamics. Negative overfitting ratios $R(L) < 0$ indicate sign reversal between IS and OOS performance, suggesting a fundamental change in market regime.

Table 10 reports the parameter-degradation results for eight moving-average horizons ($L = 30, 60, 90, 120, 180, 240, 300,$ and 360 days) evaluated over six consecutive OOS windows. The longer lookback periods (300–360 days) exhibit clear signs of overfitting with positive degradation values ranging from 0.49 to 0.76 and poor OOS performance, suggesting instability in parameter behavior. In contrast, shorter periods (30–60 days) display minimal degradation values (−0.01 to −0.14) with high consistency between IS and OOS performance, even though the 30-day period has a negative average OOS Sharpe (−0.17).

The intermediate lookback periods (90–240 days) demonstrate superior performance characteristics. Notably, the 90-day lookback offers the highest average OOS Sharpe ratio and cumulative return, while the 240-day lookback shows the most negative Sharpe degradation (−1.02), suggesting excellent generalization from IS to OOS periods. These mid-range lookbacks exhibit higher Sharpe stability values (2.29 and 1.67), indicating moderate variability across different market regimes.

Furthermore, I apply two additional validation techniques–nested cross-validation (CV) and Genetic Algorithm (GA) optimization–and compare their outcomes with the original grid-search optimization approach and traditional buy-and-hold strategies. Nested CV is particularly useful in small-sample contexts, as it robustly validates model performance and selects parameters that minimize prediction error, thus reducing the risk of overfitting (Bhattacharjee et al. 2022). Additionally, the GA method, previously employed by Allen and Karjalainen (1999) to identify profitable technical trading rules in financial markets, addresses optimization issues characterized by complex, multi-modal solution landscapes. Due to their stochastic selection and recombination processes, GAs effectively avoid local optima, offering a substantial advantage over deterministic gradient-based methods.

Table 11 presents the performance metrics of each strategy. The results indicate that all three optimization approaches outperform the buy-and-hold benchmark, confirming the robustness of the pairs trading strategy, despite the differences in the parameter selection. The nested CV reveals consistent parameter selections across folds (lookbacks of 30 and 57 days), with a low standard deviation of the Sharpe ratio (0.09), supporting the robustness of shorter lookback periods with the highest stability of the optimized parameter across different folds. For the GA optimization, I calculate the overfitting ratio ($\frac{ISperformance}{OOSperformance}$) that compares IS and OOS performance. The overfitting ratio (0.58) suggests that the parameters perform better with OOS than IS data. This low ratio provides evidence that the GA approach guards against overfitting despite the simultaneous optimization of multiple parameters.

These findings explain why different optimization approaches yield different parameters: nested CV identified shorter lookbacks as optimal because it prioritizes consistent performance across different data segments rather than maximizing absolute performance; the GA's intermediate solution (194 days) represents a reasonable compromise between performance and stability; and the original optimization found the highest IS Sharpe ratio but with a parameter choice (291 days) that our analysis reveals to be less robust across changing market conditions. Therefore, the results suggest that the strategy should avoid longer lookback periods (over 300 days) due to their tendency toward overfitting. Instead, a calibration approach that dynamically selects between shorter periods (for consistency) and intermediate periods (for performance) would allow the strategy to achieve higher Sharpe ratios while maintaining parameter stability across varying market regimes.

4.2.4 | Momentum Strategy

This section implements a momentum strategy and compares its performance with that of a passive buy-and-hold strategy. The analysis includes BTC, which is the cryptocurrency with the highest price increase in the sample, to minimize concerns of cherry-picking bias. Notably, the pairs trading strategy has thus far demonstrated its potential as a viable investment

TABLE 10 | Degradation analysis: sharpe & return stability.

Lookback (d)	Avg. OOS sharpe	Sharpe degr.	Sharpe stability	Avg. OOS cum ret	Cum ret degr.	Sharpe IS/OOS
30	−0.173	−0.007	0.900	−0.055	−0.237	1.041
60	0.208	−0.141	1.148	0.059	0.072	0.323
90	0.893	−0.683	2.291	0.211	0.165	0.236
120	0.579	−0.322	1.600	0.162	0.278	0.445
180	0.082	−0.127	1.069	−0.002	−0.062	−0.546
240	0.764	−1.020	1.668	0.119	−0.424	−0.336
300	−0.194	0.493	0.803	−0.056	0.497	−1.548
360	−0.022	0.756	0.934	−0.035	1.166	−33.744

TABLE 11 | Performance metrics comparison.

Strategy	Cumulative return	Sharpe ratio	Calmar ratio	Max drawdown
Original (LB = 291)	1.8699	2.1165	12.9830	−0.1440
Nested CV (LB = 30)	1.1767	1.5219	4.5654	−0.2577
Genetic algorithm	0.5170	1.2039	4.3326	−0.1193
LTC. USD (Buy & Hold)	−0.3106	−0.3608	0.6104	−0.5089
DOGE. USD (Buy & Hold)	0.2601	0.1221	0.5518	−0.4713

alternative, even in a trending market where strategies such as momentum and buy-and-hold tend to perform well.

The momentum strategy operates under the assumption that past returns can provide predictive insights into future price movements (Asness et al. 2013). Trading signals derive from the cumulative sum of historical log returns over a rolling window. Specifically, the momentum signal, M_t is derived by aggregating historical returns over a rolling lookback window of length L, as follows:

$$M_t = \text{sign}\left(\sum_{i=1}^{L} R_{t-i}\right), t = L + 1, \ldots, T, \tag{17}$$

where L is the lookback period (e.g., 30 trading days), $\sum_{i=1}^{L} R_{t-i}$ represents the cumulative sum of past returns over the L-day window ending at time t, and $\text{sign}(x)$ is the sign function.

Table 12 presents the performance metrics of the passive and momentum strategies for each cryptocurrency. Among the strategies, BTC exhibits the highest Sharpe ratio. However, it still underperforms when compared to the pairs trading strategy, which has thus far demonstrated superior risk-adjusted returns. In the appendix, I present Figure A1 that illustrates the performance of the momentum strategy across different lookback periods. Notably, the momentum strategy exhibits greater variability in performance across different lookback windows compared to the pairs-trading strategy.

Comparing the pairs trading strategy with BTC is crucial given BTC's predominant role in cryptocurrency markets. Koutmos (2018) demonstrates that BTC is the primary contributor to volatility and return spillovers among major cryptocurrencies by market capitalization and serves as a proxy for contagion risk. This analysis reveals that pairs trading is a viable investment approach that offers effective risk management in the cryptoasset market.

Table 13 presents the performance metrics of the pairs trading strategy relative to BTC. The strategy exhibits a statistically significant positive alpha (0.65, $p < 0.01$), indicating substantial risk-adjusted outperformance relative to the BTC benchmark. The beta coefficient (−0.13) reveals a modest negative relationship with BTC returns, suggesting potential diversification benefits. This low systematic exposure is further evidenced by the minimal R-squared value (0.027), indicating that BTC returns explain only 2.7% of the strategy's return variance (see Figure B1 for a summary of the correlation diagnostic reported in the Appendix). This statistical independence suggests the strategy captures return drivers distinct from traditional cryptocurrency market exposure. The strategy's tracking error (0.30) quantifies the volatility of excess returns relative to BTC, while the information ratio (2.18) measures the risk-adjusted excess return. This information ratio is particularly noteworthy, as it exceeds conventional thresholds for superior investment performance. These metrics collectively demonstrate the strategy's capacity to generate consistent risk-adjusted returns while maintaining low correlation with the broader cryptocurrency market. The findings diverge from those of Singh et al. (2023), who argue that dual momentum strategies outperform pairs trading in cryptoasset markets by capitalizing on upward market movements. While they suggest that momentum strategies benefit from reduced transaction costs due to lower trading frequency, this analysis demonstrates that pairs trading achieves superior risk-adjusted performance, even after accounting for trading costs and during bull market conditions. This outperformance is attributed to several key factors.

First, pairs trading demonstrates enhanced resilience during periods of high volatility, as its performance is less dependent on directional market trends. Second, the systematic optimization of the lookback window proves crucial in capturing positive returns. Third, the implementation of dynamic risk management mechanisms–specifically, the trailing stop-loss and volatility filter–effectively mitigates the impact of market turbulence and optimizes trade execution. The empirical results demonstrate that a well-calibrated pairs trading strategy can outperform traditional pairs trading, momentum and buy-and-hold strategies on a risk-adjusted basis. This finding is particularly significant because it challenges the conventional wisdom regarding the superiority of trend-following strategies in cryptocurrency markets. The strategy's effectiveness stems from

TABLE 13 | Performance analysis of pairs trading strategy versus bitcoin.

Metric	Value	Stats.
$Alpha_{annual}$	0.65	3.03***
Beta	−0.13	−3.73***
R-squared	2.75%	—
Tracking error	0.30	—
Information ratio	2.18	—

***$p < 0.01$.

TABLE 12 | Performance metrics for cryptocurrencies and momentum strategies.

Cryptos	Sharpe ratio	Calmar ratio	Max. drawdown
LTC. USD	−0.36	−0.34	0.51
DOGE. USD	0.12	0.27	0.47
BTC. USD	1.37	2.70	0.23
LTC. USD (Momentum)	−1.04	−0.69	0.83
DOGE. USD (Momentum)	0.24	0.40	0.52
BTC. USD (Momentum)	0.91	1.29	0.32

its ability to implement risk management protocols and maintain performance across varying market conditions.

4.3 | Discussion of Strategy Limitations

In this section, I acknowledge potential limitations associated with the trading strategy developed in this study. For instance, analyses of Sharpe ratios of investment strategies published in academic journals demonstrate that return predictability deteriorates significantly after publication. Falck and Rej (2022) replicate a sample of 60 published investment strategies, demonstrating that Sharpe ratios decline by approximately half following publication. Mclean and Pontiff (2016) explore potential explanations for this decline. If return predictability reflects rational expectations given the risk factors (such as systematic risk, transaction costs, or taxes), rational investors should already be accounting for these factors in their decision-making processes (Cochrane 1999). Consequently, the publication of these predictors would not provide new information that could alter their behavior. This perspective aligns with the rational expectations hypothesis introduced by Muth (1961), which postulates that expectations coincide with optimal forecasts derived from all available information. Under rational expectations, returns before and after publication should remain consistent after accounting for statistical biases. In contrast, if return predictability arises from market mispricing, publication could attract sophisticated investors who act on this information, thereby reducing or even eliminating the mispricing. However, mispricing may not fully disappear due to frictions that prevent arbitrage from fully correcting prices, such as systematic noise trader risk, idiosyncratic risk, and transaction costs.

Another potential limitation concerns the fundamental premise of the pairs-trading approach, which exploits mean-reversion properties by capitalizing on temporary mispricings of assets whose prices are expected to revert to a long-term equilibrium. Such strategies are inherently dependent upon the statistical concept of cointegration (Elliott and * 2005). Keilbar and Zhang (2021), who examine cointegration dynamics among cryptocurrencies, find stronger nonlinear relationships during market stress periods. Their analyses indicate that large deviations from long-run equilibria offer profitable opportunities, whereas OOS performance requires more cautious interpretation, with success depending on the long-term stability of equilibrium relationships. Hence, the careful pre-selection of trading pairs based on cointegration relationships is essential. However, given the heterogeneous and dynamic nature of cryptocurrency markets, the effectiveness of such relationships may vary over time as cointegration may weaken or even disappear. Despite these challenges, the proposed strategy consistently outperforms simpler pairs-trading benchmarks both IS and OOS.

While no trading approach can guarantee future success despite extensive robustness testing, this manuscript makes several novel contributions to the understanding of empirical investment strategies, particularly those used in the highly uncertain cryptocurrency market. First, it demonstrates the critical importance of optimizing the lookback period. Although the study identified time-varying optimal parameters, the proposed strategy exhibits significantly less performance variation than the widely studied momentum approach highlighted in Appendix A. Second, the innovative combination of volatility filtering and dynamic trailing stop-loss mechanisms reduced maximum drawdown, yielding superior risk-adjusted returns compared not only to conventional pairs-trading implementations but also to momentum and buy-and-hold strategies. Third, the strategy maintained positive performance across both bullish and bearish market regimes, demonstrating adaptability to changing market conditions.

5 | Conclusion

This study assesses the effectiveness of a cointegrated pairs-trading strategy in cryptocurrency markets by benchmarking its performance against traditional cointegration-based pairs trading and a passive buy-and-hold approach across ten major cryptocurrencies, explicitly considering bull and bear market regimes. The proposed methodology improves the baseline pairs-trading framework through systematic lookback period optimization via grid-search and incorporates adaptive trailing stop-loss and volatility filtering mechanisms to reduce downside risk. Empirical results show that the optimized pairs-trading strategy consistently outperforms traditional pairs trading, passive buy-and-hold, and momentum-based strategies across risk-adjusted metrics, including higher annualized Sharpe ratios and lower maximum drawdowns. Notably, the strategy's returns exhibit minimal correlation with broader cryptocurrency market movements, emphasizing its value as a robust diversification tool in volatile market settings.

These findings have important implications for both practice and theory. This study offers practical value for institutional investors, hedge funds, and traders seeking portfolio diversification in cryptocurrency markets. The demonstrated capacity of the risk-adjusted pairs trading strategy to generate returns independently of market direction while maintaining robust drawdown control indicates this strategy is a compelling alternative to traditional directional strategies. Furthermore, this study advances the methodological literature on trading strategy development by introducing systematic approaches for parameter optimization and risk management. However, while our empirical findings are robust, further theoretical work is needed to develop a more comprehensive framework for parameter optimization in pairs trading strategies.

Notably, the methodological framework developed in this paper provides opportunities for future research that could enhance its practical applicability and generalizability along several dimensions. First, future research could address potential constraints associated with liquidity. The cryptocurrencies selected were chosen not only based on consensus mechanisms but also on their market capitalization. Nevertheless, smaller cryptocurrencies exhibit narrower market depth, wider bid–ask spreads, and slippage, potentially increasing transaction costs and posing execution risks (Brown et al. 2009). For instance, Tripathi and Dixit (2022) underscore how liquidity commonality exacerbates transaction costs during market-wide liquidity events, significantly impairing execution quality and profitability. Such liquidity-induced volatility constrains strategy implementation because it increases the likelihood of extreme market movements, significantly elevating real-world execution challenges and transaction costs beyond typical assumptions. This aligns with Tang and Wang (2022)'s empirical demonstration that liquidity shocks predict higher long-term volatility and frequent large crashes in cryptocurrency markets, necessitating rigorous risk-management strategies to control execution costs and maintain profitability.

Second, future research should evaluate the implications of scalability for practical strategy execution. I If the proposed strategy were widely adopted, the mean-reversion properties driving profitability might weaken, leading to crowding-out effects, as highlighted by Mclean and Pontiff (2016). Additionally, optimal position sizing likely produces diminishing returns as capital allocation surpasses critical thresholds. As Barras et al. (2022) illustrate, increased fund size negatively impacts strategy performance, primarily due to diseconomies of scale arising from trading frictions, higher transaction costs, execution difficulties at scale, and a general dilution of skill and investment opportunities–all of which may be significantly affected by liquidity-related market impacts. Yan (2008) provides empirical evidence that liquidity exacerbates the negative relationship between fund size and fund performance, particularly among growth-oriented and high-turnover funds with substantial immediacy demands.

Finally, while the current analysis provides insights for pairs trading among established cryptocurrencies, broader validation across diverse token universes would enhance the robustness and generalizability of the findings. Moreover, the methodological framework presented in this study has potential applicability beyond cryptocurrencies, extending to various other asset classes.

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Conflicts of Interest

The author decares no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in: Baptista Palazzi, Rafael (2024), “Trading Games: Beating Passive Strategies in the Bullish Crypto Market”, Mendeley Data, V1, !https://doi.org/10.17632/2kky7c6xkn.1.

Endnotes

[1] Out of 37 cointegrated pairs, 13 (approximately 35%) exhibited a positive return during the OOS analysis. These results are not reported here but are available upon request.

[2] XRP was initially included as a representative asset. Although Ripple's RPCA differs from standard PoS, it is conceptually closer to PoS than PoW because it validates transactions through consensus among trusted validators selected based on reputation and reliability. In contrast, PoW relies explicitly on energy-intensive computational mining. Given the primary aim of distinguishing traditional PoW from newer consensus approaches, RPCA was grouped alongside PoS-like mechanisms.

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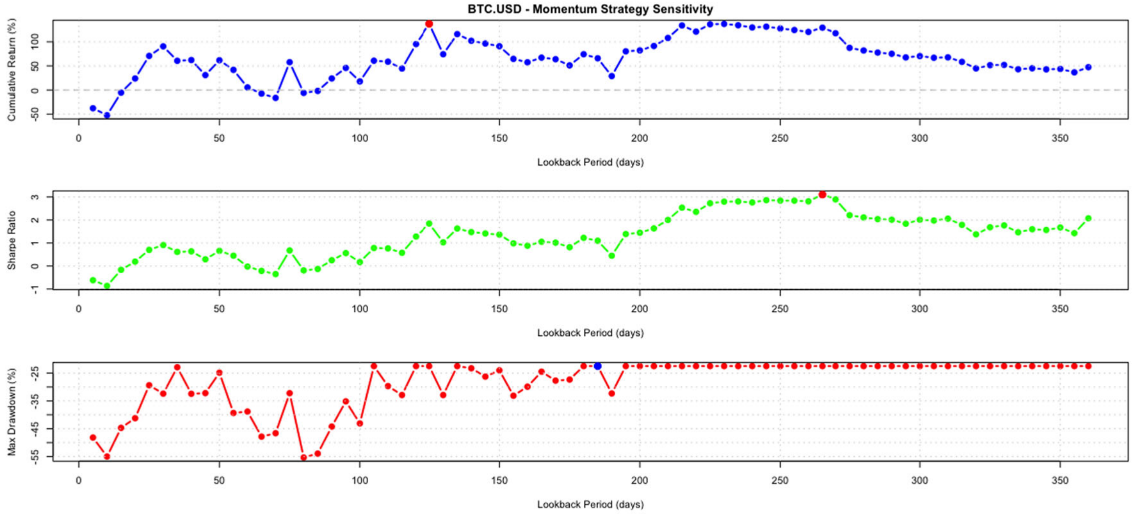
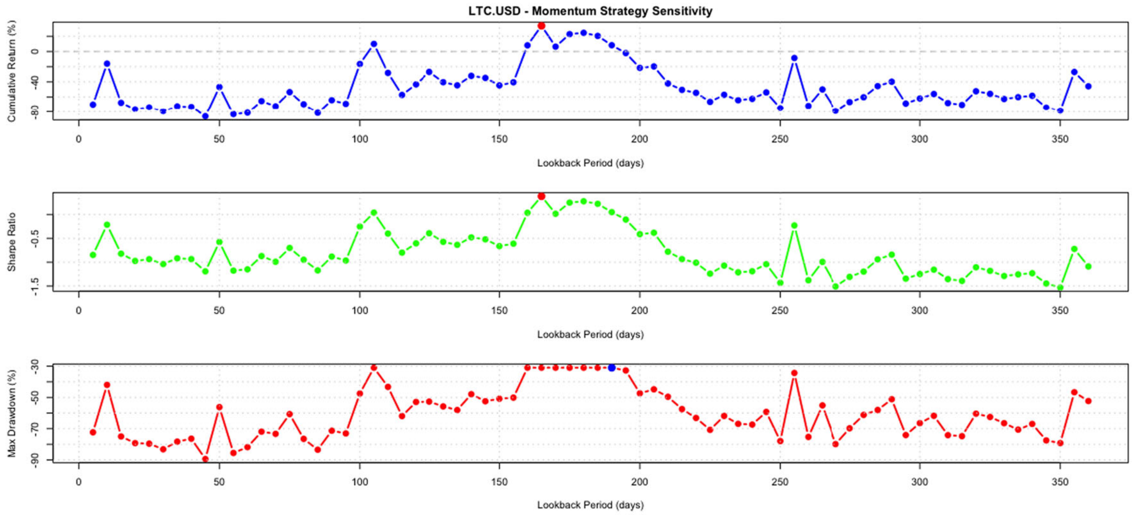
Appendix A

Momentum Lookback Analysis

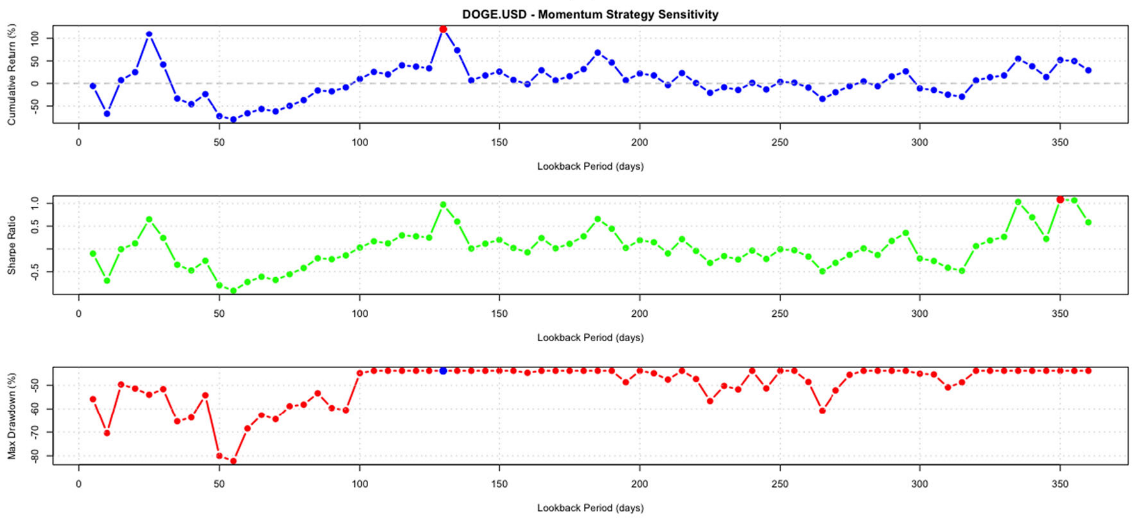
Figure A1 shows the momentum strategy's performance across lookback periods. For Litecoin (LTC), returns range from −0.86 to 0.34, with the optimal performance occurring at a 165-day lookback. Dogecoin (DOGE) is even more sensitive to lookback, with returns spanning from −0.80 to 1.22; the best performance is observed with a 130-day lookback, while the worst occurs with a 55-day lookback. Bitcoin (BTC) returns vary between −0.52 and 1.37, with the optimal period lying between 125 and 230 days, suggesting that medium-term trends are most effectively captured. All three cryptocurrencies tend to perform better with medium-term lookback periods (100–250 days) rather than shorter windows, implying that price trends persist over multi-month horizons, even though the ideal window differs by asset. This marked variability underscores the critical role of lookback in strategy success. In contrast, the pairs-trading strategy (see Figure 5) exhibits more robust performance across a range of lookback periods, consistently generating positive returns from 0.87 to 2.13.



(a)



(b)



(c)

FIGURE A1 | Performance of the momentum strategy (cumulative return, sharpe ratio, and maximum drawdow) for BTC, LTC, and DOGE across different lookback periods. Each plot illustrates how the cumulative returns vary with the choice of lookback window for each cryptocurrency. (a) Momentum strategy performance for BTC. USD across varying lookback periods. (b) Momentum strategy performance for LTC. USD across varying lookback periods. (c) Momentum strategy performance for DOGE. USD across varying lookback periods.

Appendix B

Diagnostic Plots

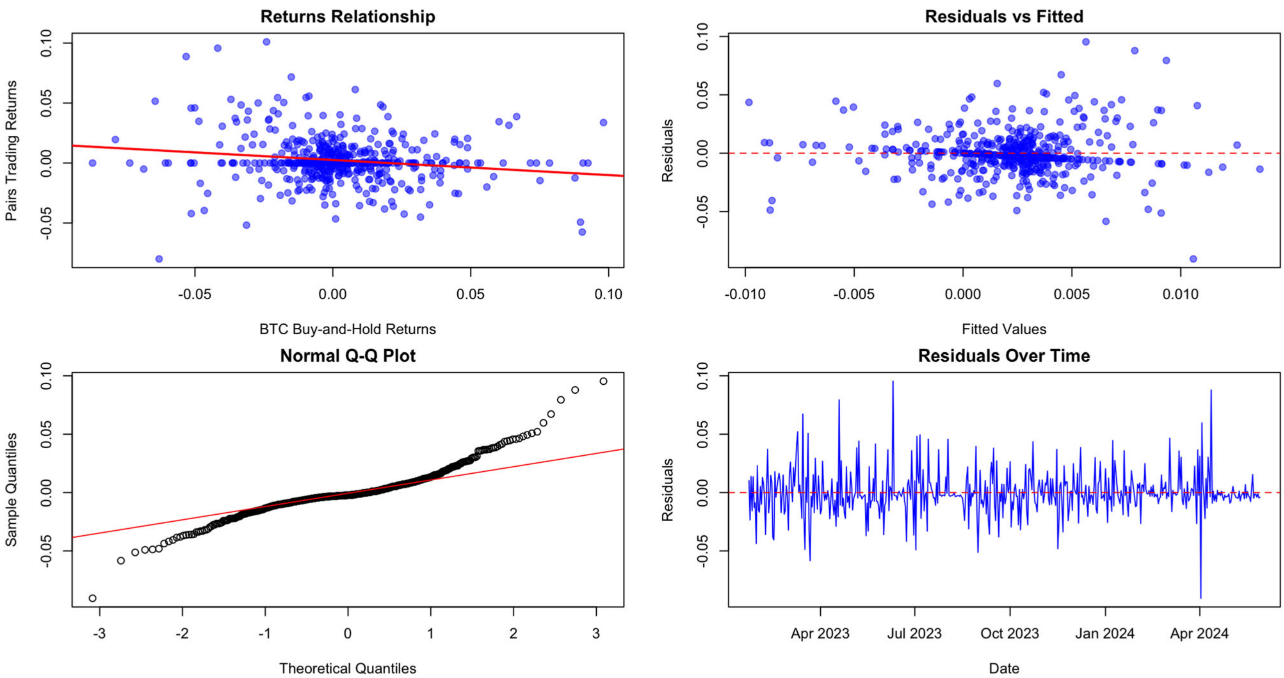


FIGURE B1 | Diagnostic Plots for Pairs Trading Strategy Performance Analysis. *Note:* This figure provides four diagnostic plots examining the relationship between pairs trading returns and Bitcoin returns. Panel A (top left) illustrates the relationship between returns, with the regression line indicating a slight negative correlation. Panel B (top right) shows residuals versus fitted values, confirming the homoscedasticity of the model. Panel C (bottom left) displays the normal Q-Q plot of residuals, which indicates approximate normality with some deviation in the tails. Panel D (bottom right) presents the time series of residuals, demonstrating consistent variance over time. The analysis period covers January 2023 to May 2024.

Appendix C

Consensus Mechanism Expanded

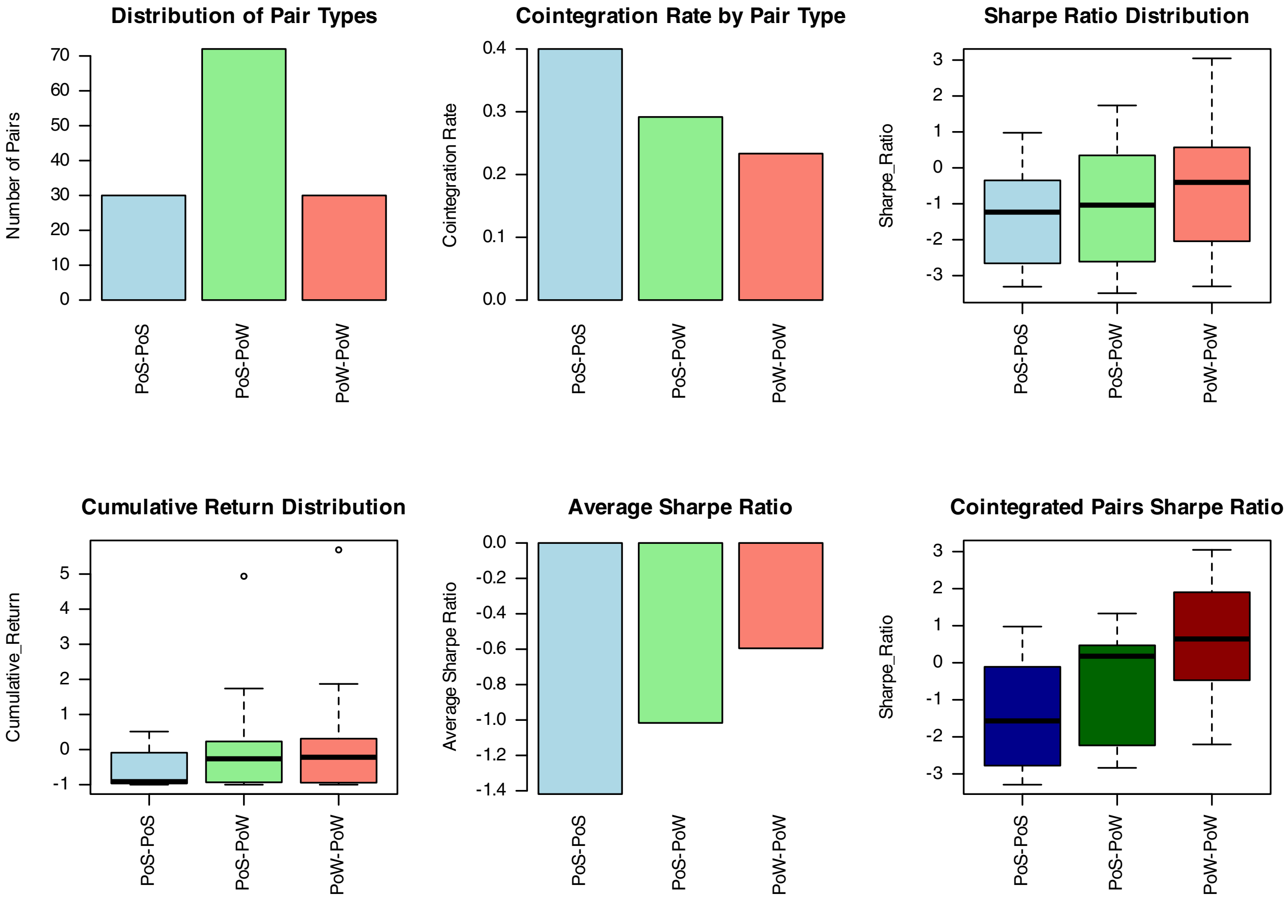


FIGURE C1 | Performance Metrics by Consensus Mechanism. Note: This figure provides the analysis of cryptocurrency pairs (total $n = 132$) categorized by their consensus mechanisms: Proof-of-Stake (PoS), Proof-of-Work (PoW), and mixed pairs (PoS–PoW). Panel (a) illustrates the overall distribution of pair types. Panel (b) compares cointegration rates across the different pair categories. Panels (c) and (d) display boxplots of Sharpe ratios and cumulative returns, respectively, highlighting variability in pair performance. Panel (e) summarizes the average Sharpe ratios by consensus category, emphasizing risk-adjusted performance differences among consensus mechanisms.

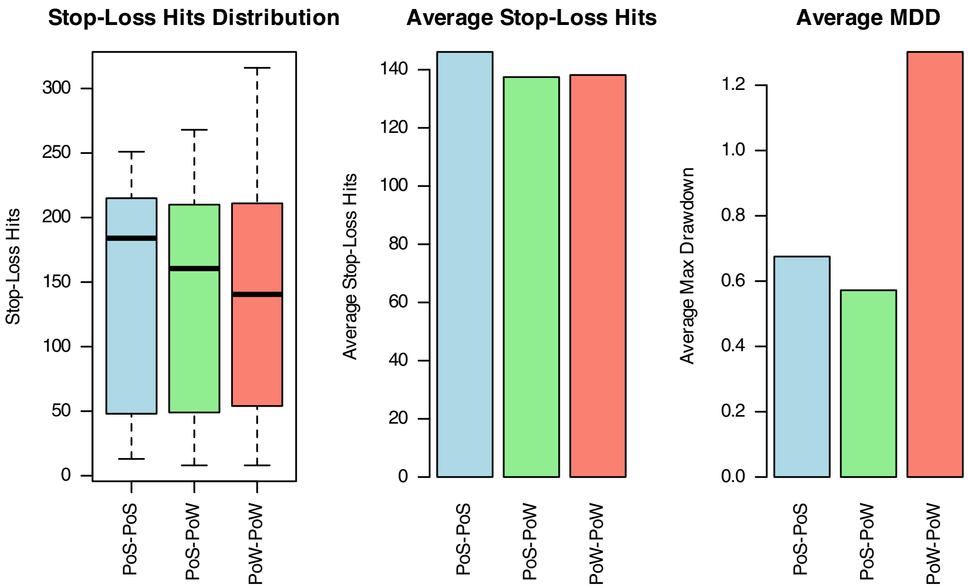


FIGURE C2 | Performance Risk Metrics by Consensus Mechanism. *Note:* This figure presents a comparative risk analysis of cryptocurrency pairs, grouped by their consensus mechanisms: Proof-of-Stake (PoS), Proof-of-Work (PoW), and mixed pairs (PoS–PoW). (1) Distribution of stop-loss hits; (2) maximum drawdown (MDD); and (3) risk-adjusted metrics.

TABLE C1 | Statistical tests of sharpe ratio and cointegration by pair type.

Test	Contrast/Model	Statistic	df	*p* value
ANOVA (homoscedastic)	Sharpe ~ Pair_Type	$F = 2.19$	2, 127	0.116
Welch ANOVA (HC)	Sharpe ~ Pair_Type, unequal var.	$F = 2.41$	2, 62	0.098[a]
Tukey HSD	PoW-PoW − PoS-PoS	$\Delta = 0.823$	—	0.096[a]
HC3 pairwise (Bonferroni)	PoS-PoS − PoW-PoW	$\Delta = -0.823$	127	0.099[a]
Chi-square	Pair_Type × Cointegrated	$\chi^2 = 2.07$	2	0.355

[a]Marginally significant at the 10% level; none of the tests reach the conventional 5% threshold.